

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 50
timesteps_orig: 60
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.1
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
group_msec: 50
samples_per_sec: 20
forecast_horizon_sec: 2
forecast_steps: 40
timesteps: 60

LSTM Multi-Step Training Report

Report Generated: 2025-05-10 17:46:25

Training Start: 2025-05-10 17-18-01

Training End: 2025-05-10 17-38-14

Training Duration: 20.21 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 48203.3813

MSE: 3774343747.8664

RMSE: 61435.6879

R2 Score: 0.7770

Multi-Step Forecast:

Forecast Steps: 40

MAE: 0.0183

MSE: 0.0005

RMSE: 0.0231

R2 Score: 0.9433

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 20,517 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

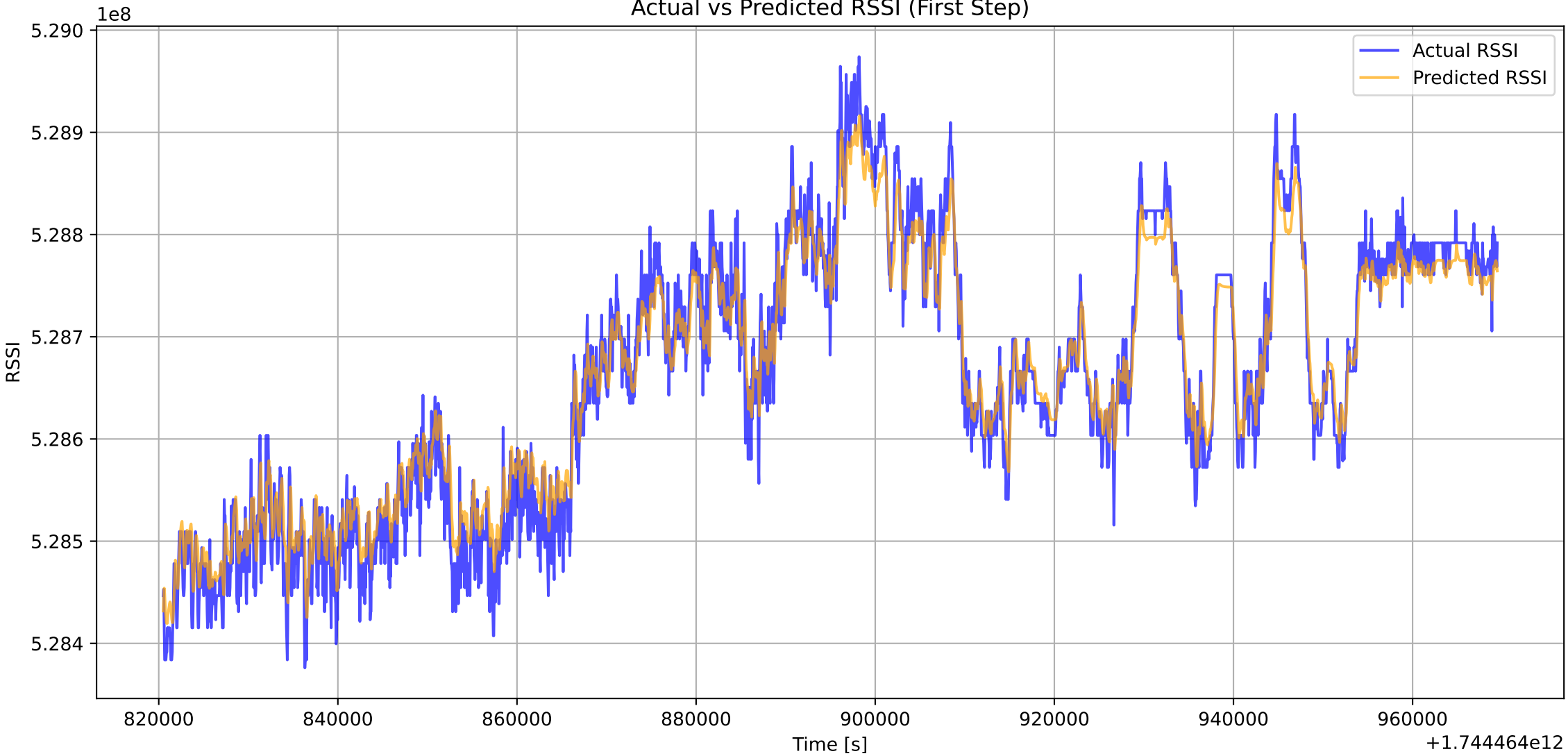
1744397085760
1744397085801
1744397085856
1744397086014
1744397086050

...

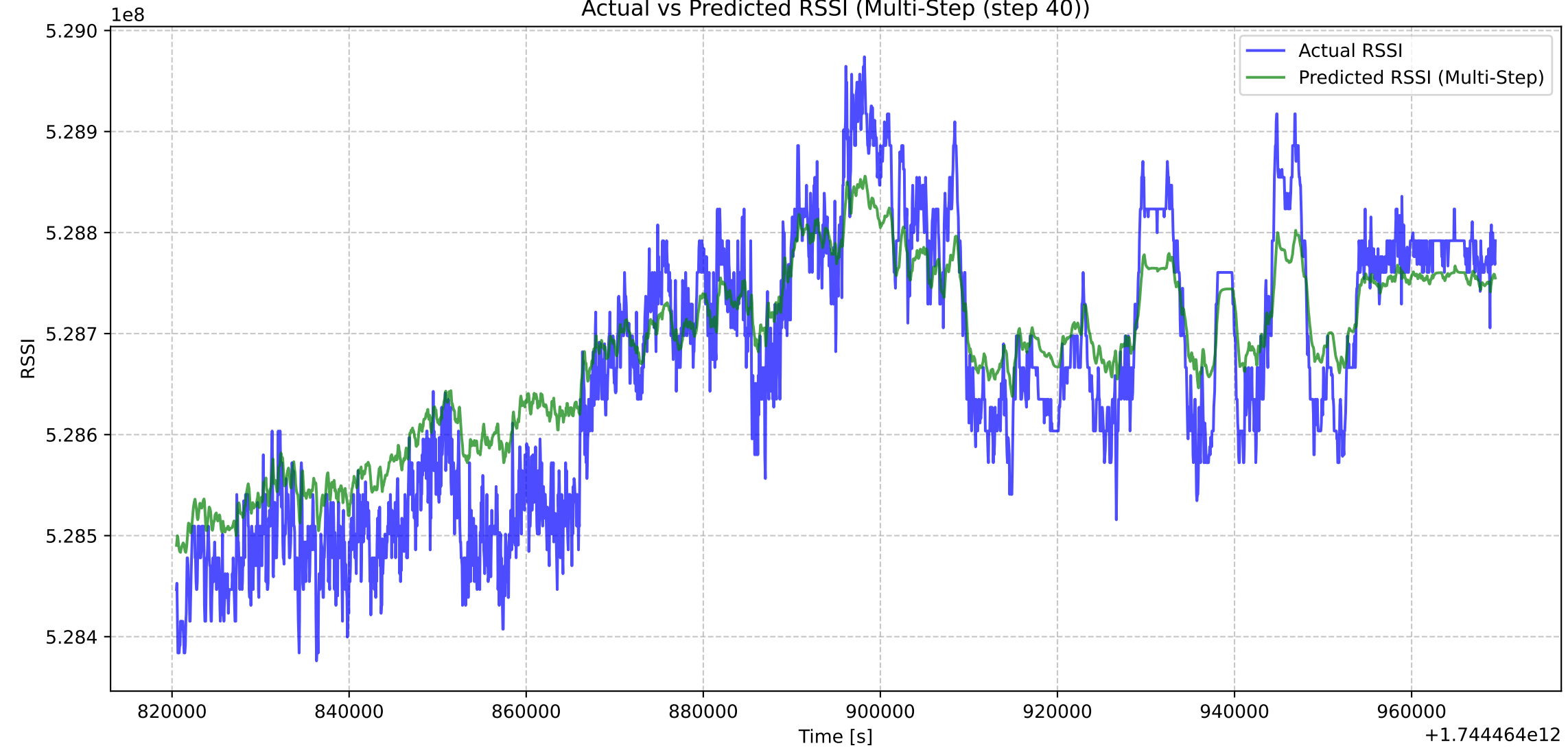
Last 5 timestamps AFTER:

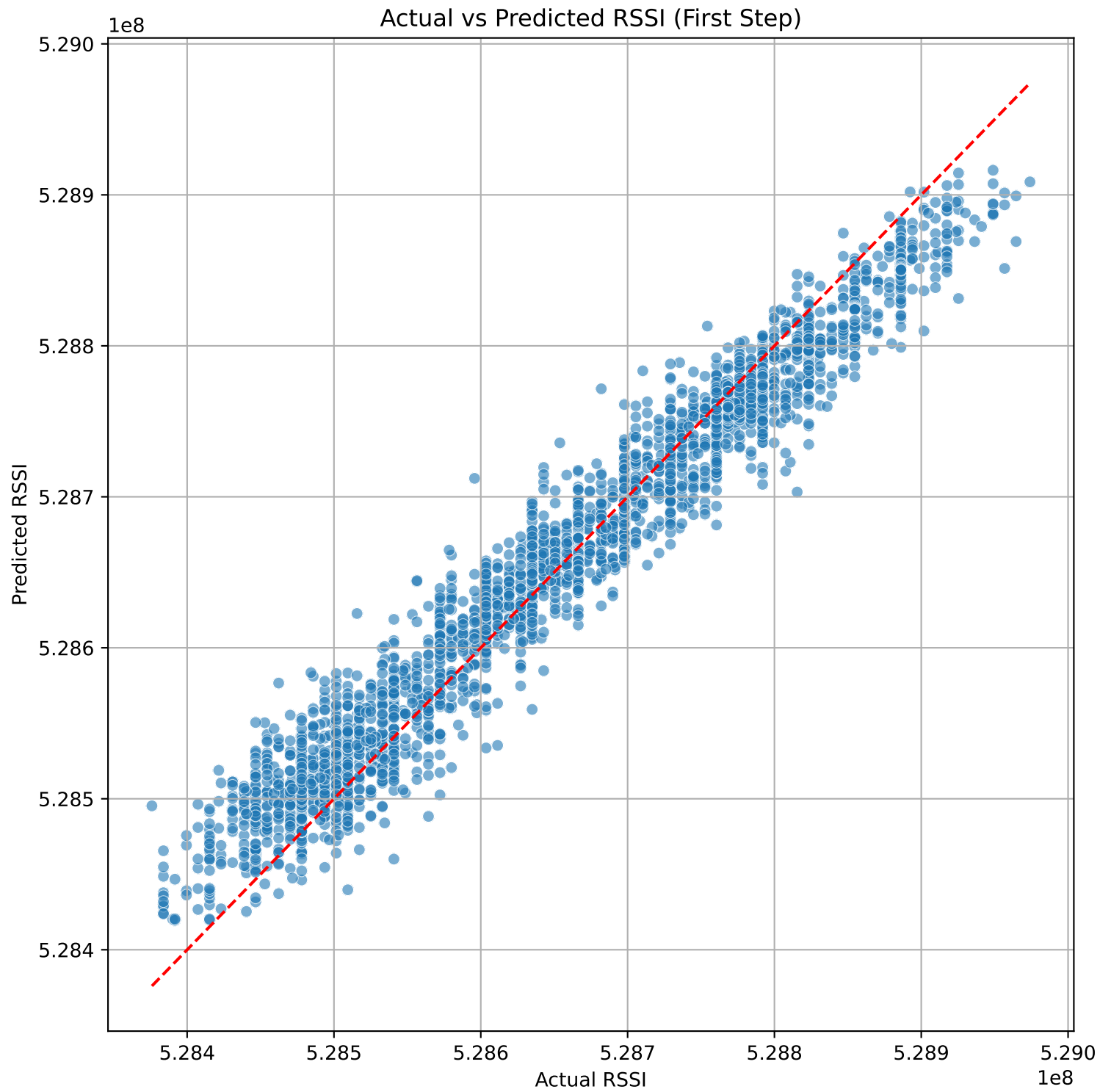
1744464971209
1744464971257
1744464971306
1744464971353
1744464971402

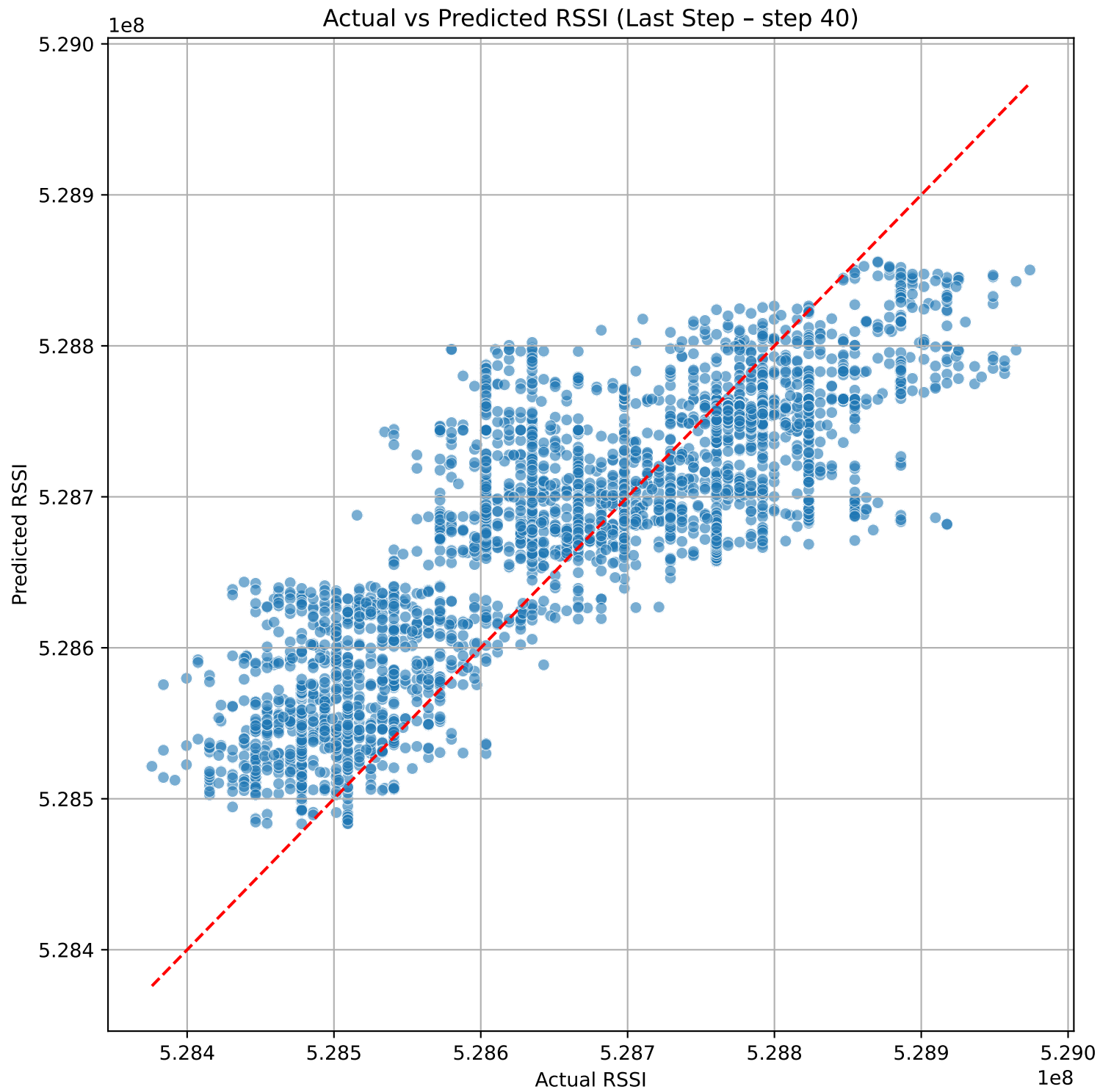
Actual vs Predicted RSSI (First Step)



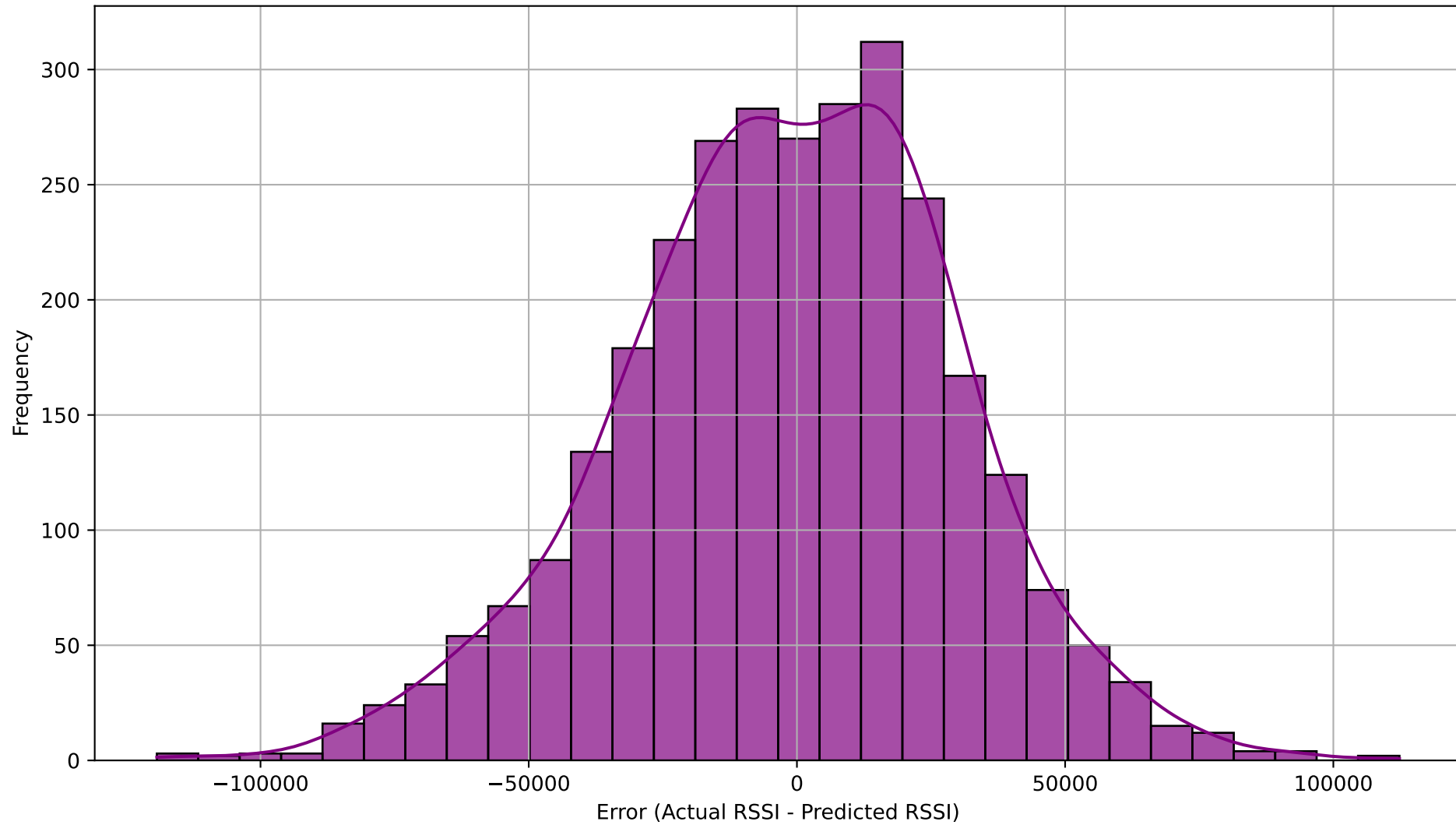
Actual vs Predicted RSSI (Multi-Step (step 40))



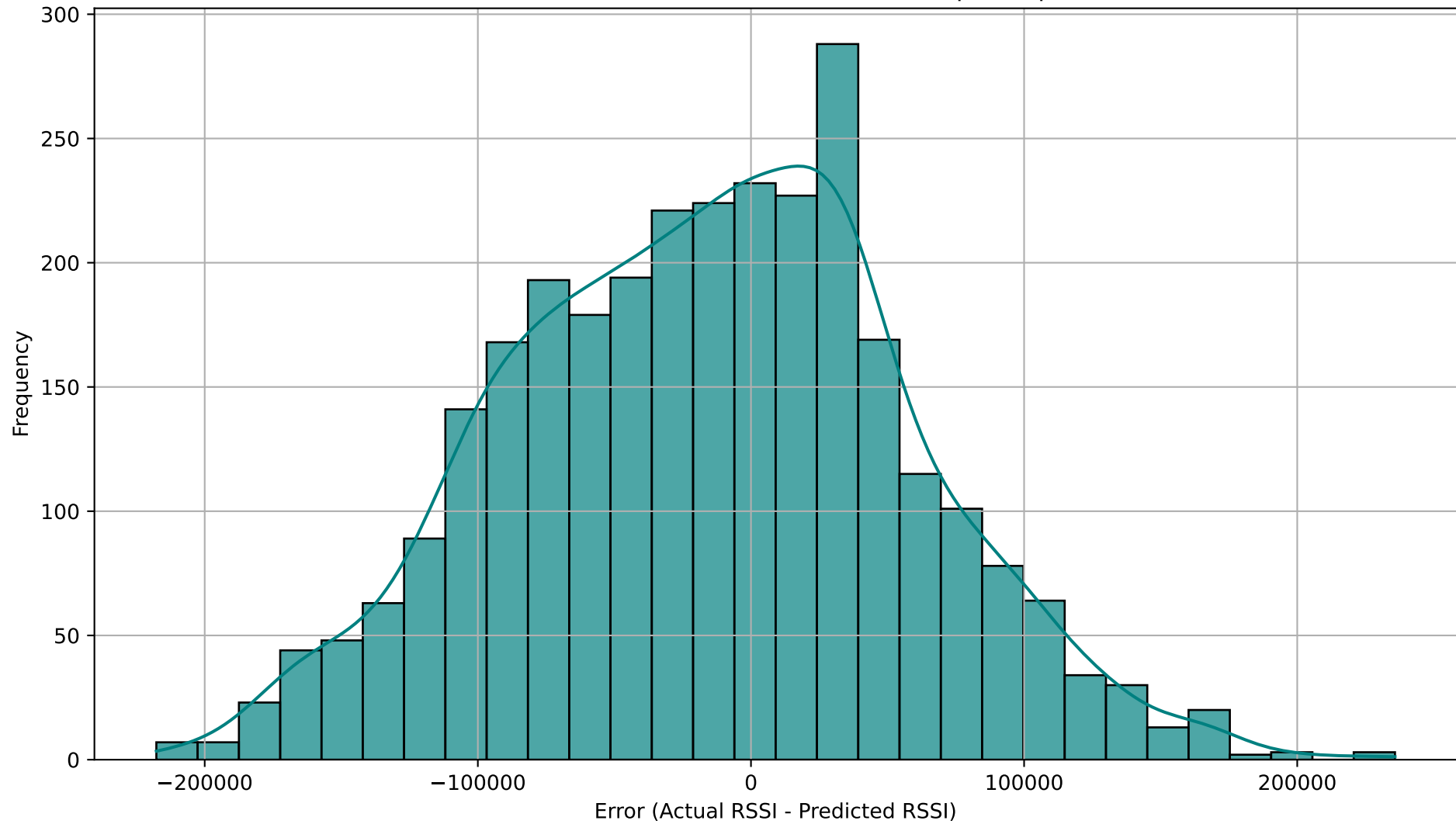




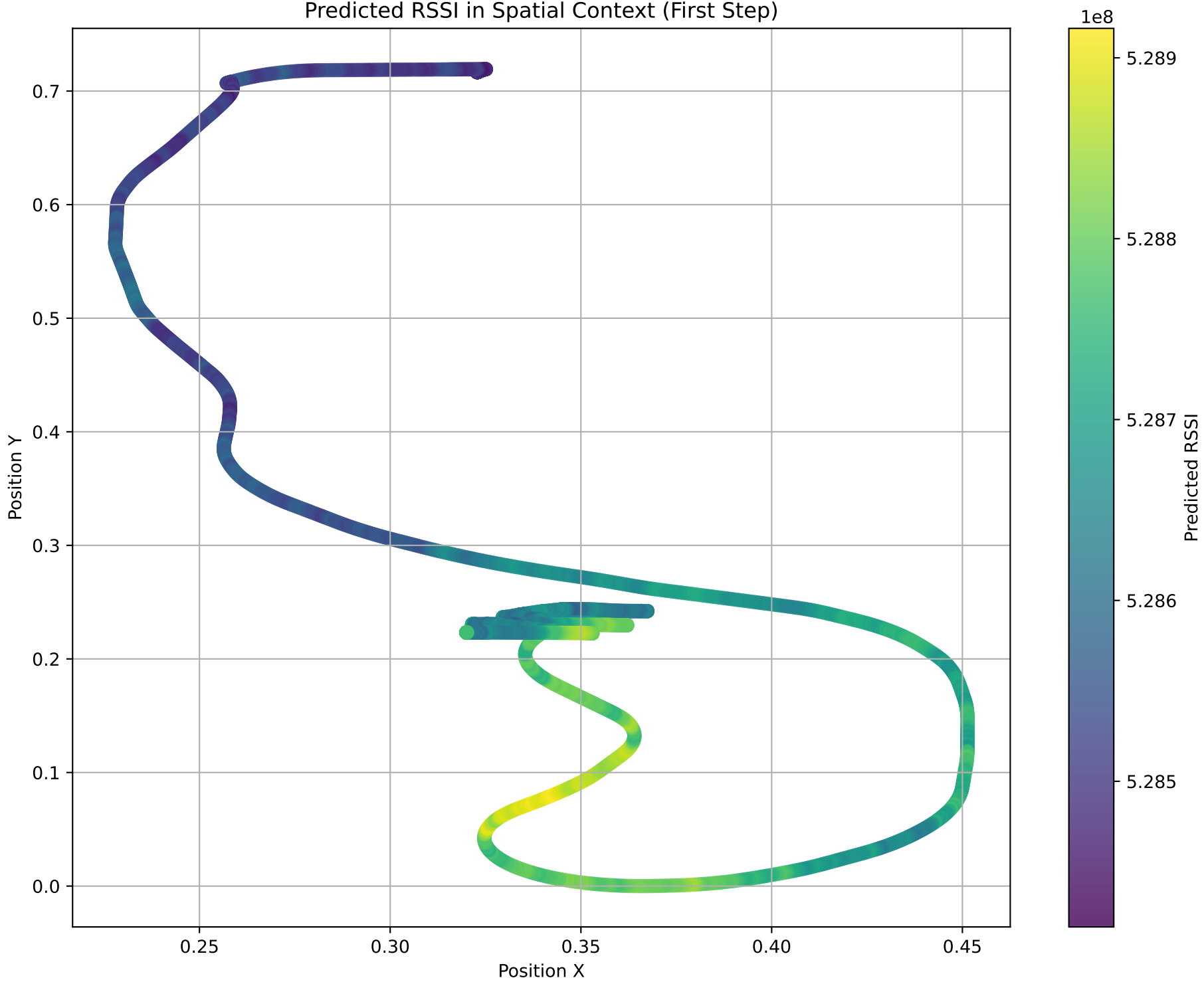
Distribution of Prediction Errors (First Step)



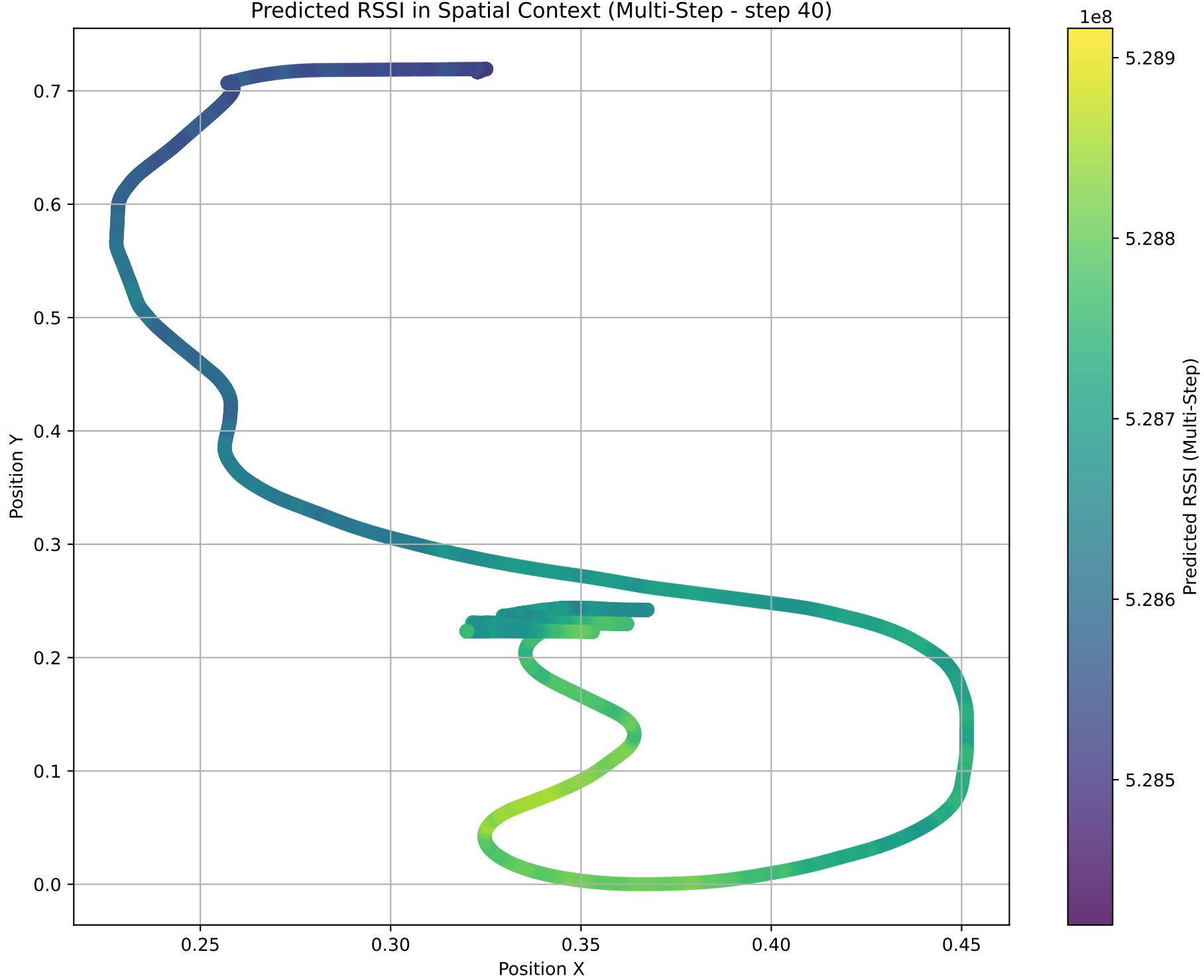
Distribution of Prediction Errors (Last Step - step 40)



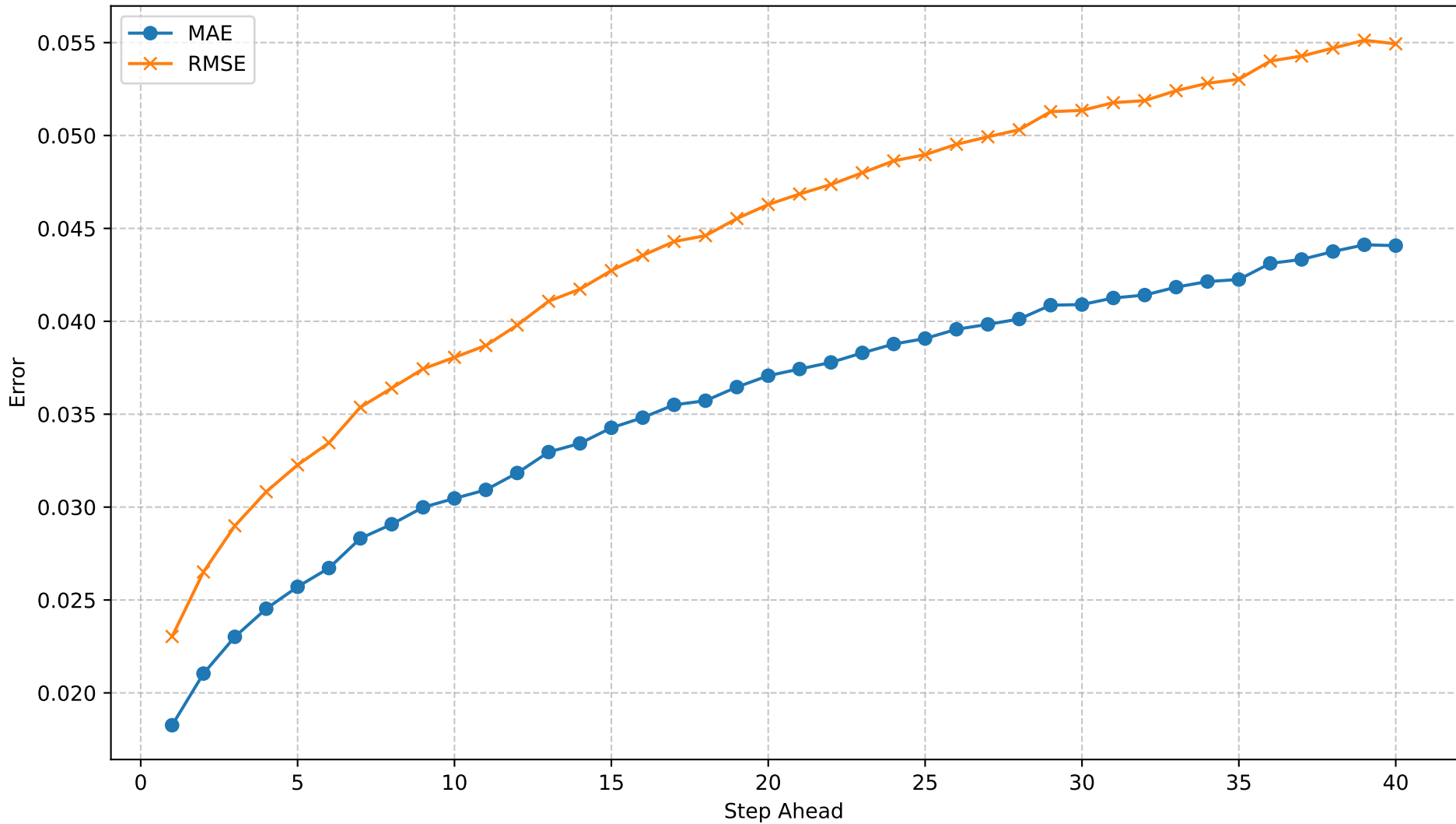
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 40)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 40

MAE / RMSE (dB)

- 1º passo : 0.02 / 0.02
- Mediana : 0.04 / 0.05
- 40º passo : 0.04 / 0.05

Crescimento do erro: 141.4% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 39 (0.04), 40 (0.04), 38 (0.04)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.